

Tennis_Points

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Data: https://www.tennisabstract.com/charting/20250409-M-Monte__Carlo__Masters-R32-Francisco__Cerundolo-Carlos__Alcaraz.html

```
tennis_data <- read.table("STATS_Alcaraz_Cerundelo.txt", header = TRUE, sep = "\t",
  stringsAsFactors = FALSE, fill = TRUE)
```

Format Table to be easy to work with

```
# Remove rows with "SET" in the STATS.OVERVIEW column
tennis_data_formatted <- tennis_data[!grepl("SET", tennis_data$STATS.OVERVIEW), ]

# Convert percentage columns to numeric
percentage_cols <- c("A.", "DF.", "X1stIn", "X1st.", "X2nd.", "RPW.")
tennis_data_formatted[percentage_cols] <- lapply(tennis_data_formatted[percentage_cols],
  function(x) as.numeric(gsub("%", "", x)) / 100)

# Extract FH/BH values from Winners and UFE columns
tennis_data_formatted$Winners_FH <- as.numeric(gsub(".*\\((\\d+)/\\d+\\.\\)",
  "\\1", tennis_data_formatted$Winners..FH.BH.))
tennis_data_formatted$Winners_BH <- as.numeric(gsub(".*\\((\\d+)/\\d+\\.\\)",
  "\\1", tennis_data_formatted$Winners..FH.BH.))
tennis_data_formatted$UFE_FH <- as.numeric(gsub(".*\\((\\d+)/\\d+\\.\\)",
  "\\1", tennis_data_formatted$UFE..FH.BH.))
tennis_data_formatted$UFE_BH <- as.numeric(gsub(".*\\((\\d+)/\\d+\\.\\)",
  "\\1", tennis_data_formatted$UFE..FH.BH.))

tennis_data_formatted <- tennis_data_formatted[,
  !names(tennis_data_formatted) %in%
  c("Winners..FH.BH.", "UFE..FH.BH.")]

colnames(tennis_data_formatted) <- c("Player", "A_pct", "DF_pct", "FirstIn", "First_pct",
  "Second_pct", "BPSaved", "RPW_pct", "Winners_FH",
  "Winners_BH", "UFE_FH", "UFE_BH")

# Format table
tennis_data_formatted$Set <- NA
tennis_data_formatted$Set[1:2] <- "Overview"
```

```

players_per_set <- 2

for (i in 3:nrow(tennis_data_formatted)) {
  tennis_data_formatted$Set[i] <- ceiling((i - 2) / players_per_set)
}
print(tennis_data_formatted)

```

##		Player	A_pct	DF_pct	FirstIn	First_pct	Second_pct	BPSaved
## 1	Francisco	Cerundolo	0.036	0.091	0.600	0.515	0.409	2/8
## 2	Carlos	Alcaraz	0.013	0.039	0.539	0.683	0.543	5/7
## 4	Francisco	Cerundolo	0.080	0.000	0.640	0.813	0.556	0/1
## 5	Carlos	Alcaraz	0.042	0.000	0.583	0.571	0.500	1/3
## 7	Francisco	Cerundolo	0.000	0.071	0.714	0.100	0.250	2/5
## 8	Carlos	Alcaraz	0.000	0.036	0.571	0.625	0.583	4/4
## 10	Francisco	Cerundolo	0.000	0.250	0.438	0.429	0.333	0/2
## 11	Carlos	Alcaraz	0.000	0.083	0.458	0.909	0.538	0/0

##	RPW_pct	Winners_FH	Winners_BH	UFE_FH	UFE_BH	Set
## 1	0.382	4	5	10	16	Overview
## 2	0.527	13	5	13	16	Overview
## 4	0.458	3	2	4	2	1
## 5	0.280	5	0	5	9	1
## 7	0.393	1	1	3	5	2
## 8	0.857	5	2	4	3	2
## 10	0.292	0	2	3	9	3
## 11	0.625	3	3	4	4	3

Define loop to create plots for visualizing momentum shifts

```

# Create an empty list to store the plots
plot_list <- list()
set_data <- tennis_data_formatted[tennis_data_formatted$Set != "Overview", ]
stats_to_analyze <- c("First_pct", "Second_pct", "Winners_FH", "Winners_BH",
  "UFE_FH", "UFE_BH", "RPW_pct")

# Function to calculate rolling average
calculate_rolling_avg <- function(x) {
  rollapply(x, width = 3, FUN = mean, partial = TRUE, align = "center")}

# Loop through stats, calculate rolling averages, create plots, and store them in the list
for (stat in stats_to_analyze) {
  set_data[, paste0(stat, "_rolling")] <- ave(set_data[, stat], set_data$Player, set_data$Set, FUN = calculate_rolling_avg)

  # Create plot for current stat with updated labels and title
  plot <- ggplot(set_data, aes(x = Set, y = .data[[paste0(stat, "_rolling")]]),
    color = Player, group = Player) +

    geom_line() +
    geom_point() +
    labs(title = paste("",
      ifelse(stat == "First_pct", "1st Serve Percentage",
        ifelse(stat == "Second_pct", "1st Serve Percentage",
          ifelse(stat == "Winners_FH", "Forehand Winners",

```

```

        ifelse(stat == "Winners_BH", "Backhand Winners",
              ifelse(stat == "UFE_FH", "Forehand Unforced Errors",
                    ifelse(stat == "UFE_BH", "Backhand Unforced Errors",
                          ifelse(stat == "RPW_pct", "Return Points",
                                ifelse(stat == "Winners_FH", "Stroke Winners",
                                      ifelse(stat == "Winners_BH", "Stroke Winners",
                                            ifelse(stat == "UFE_FH", "Stroke Errors",
                                                  ifelse(stat == "UFE_BH", "Stroke Errors",
                                                        ifelse(stat == "RPW_pct", "Percentage", stat)))))))))
    "(Rolling Average)" +
    theme_minimal()

    # Store the plot in the list
    plot_list[[stat]] <- plot
  }
}

```

Use Patchwork to combine plots

```

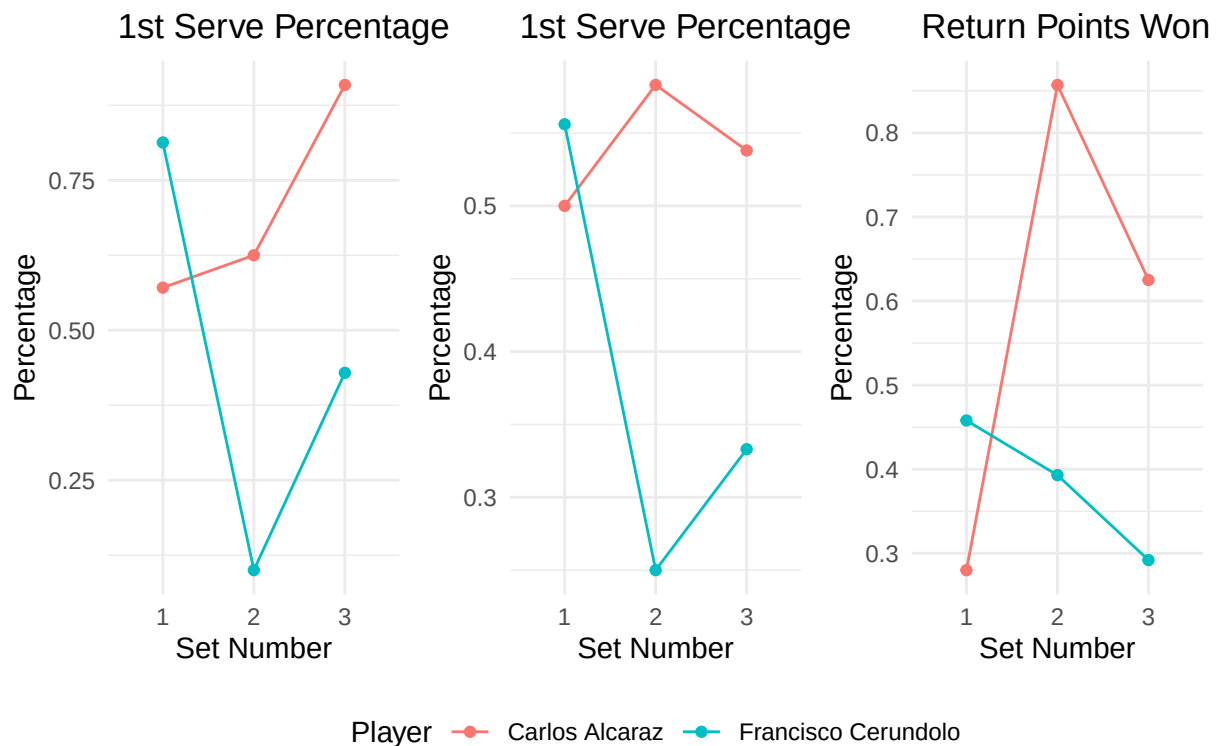
# Create the first combined plot
combined_plot1 <- (plot_list[[1]] + plot_list[[2]] + plot_list[[7]] +
  plot_layout(guides = "collect") & theme(legend.position = "bottom")) +
  plot_annotation(title = element_text("Moving Averages of Serve Stats"),
    theme = theme(plot.title = element_text(size = 14)))

# Create the second combined plot
combined_plot2 <- (plot_list[[4]] + plot_list[[5]] + plot_list[[6]] + plot_list[[3]] +
  plot_layout(guides = "collect") & theme(legend.position = "bottom")) +
  plot_annotation(title = element_text("Moving Averages of Ground Strokes"),
    theme = theme(plot.title = element_text(size = 14)))

combined_plot1

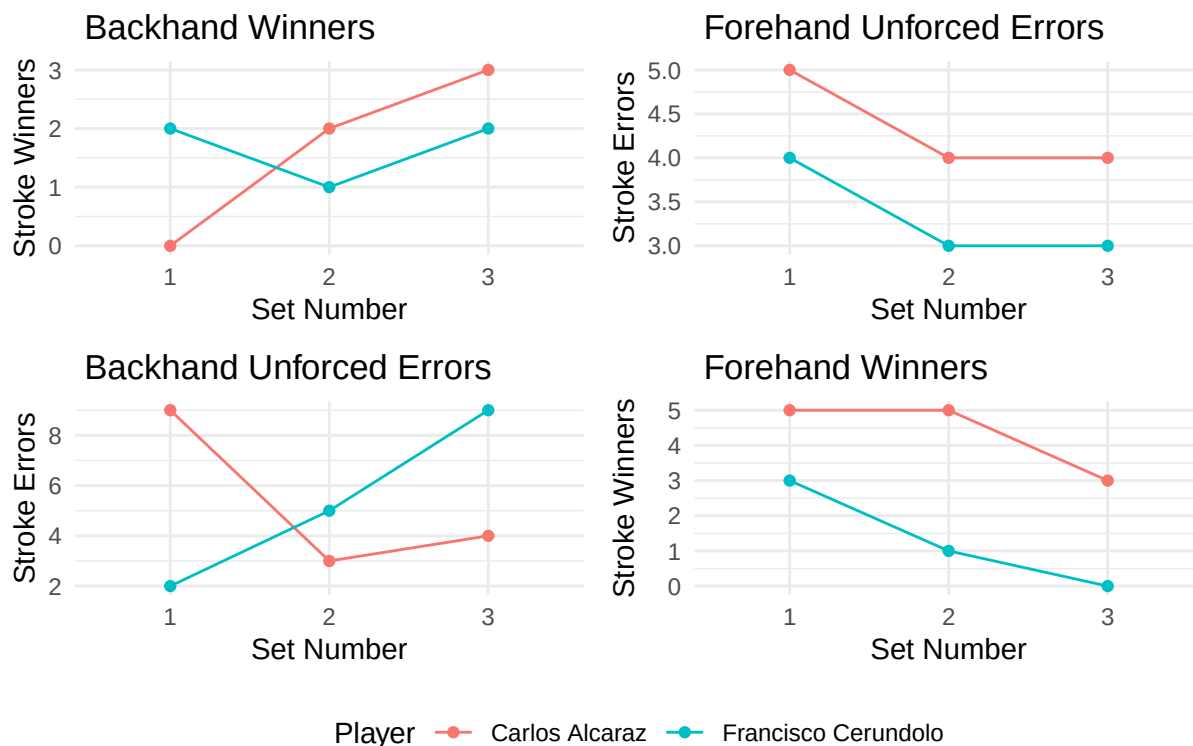
```

Moving Averages of Serve Stats



combined_plot2

Moving Averages of Ground Strokes



Create Correlation Matrix to see which variables are important

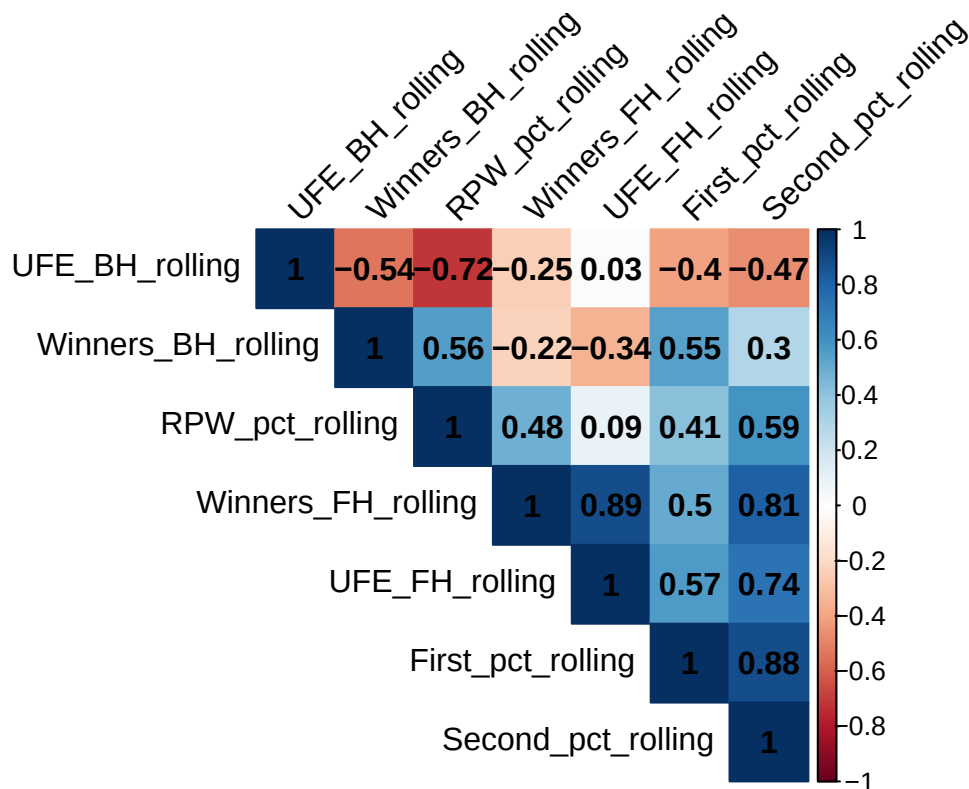
```
rolling_avg_cols <- paste0(stats_to_analyze, "_rolling")
correlation_data <- set_data[, rolling_avg_cols]
correlation_matrix <- cor(correlation_data, use = "pairwise.complete.obs")

# Calculate p-values
p_values <- matrix(NA, nrow = ncol(correlation_matrix), ncol = ncol(correlation_matrix))
for (i in 1:ncol(correlation_matrix)) {
  for (j in 1:ncol(correlation_matrix)) {
    test_result <- cor.test(correlation_data[, i], correlation_data[, j])
    p_values[i, j] <- test_result$p.value
  }
}

rownames(p_values) <- colnames(correlation_matrix)
colnames(p_values) <- colnames(correlation_matrix)

# Visualize entire correlation matrix
corrplot(correlation_matrix, method = "color", type = "upper",
          order = "hclust", addCoef.col = "black", tl.col = "black", tl.srt = 45,
          title = "Correlation Matrix of Rolling Averages", mar = c(0, 0, 2, 0))
```

Correlation Matrix of Rolling Averages



```
# Add this line to see only p values
# mar = c(0, 0, 2, 0), p.mat = p_values, sig.level = 0.05, insig = "blank")
```

Interpretation

- This script can be used to identify momentum changes in serve and ground stroke for player data

Strong Positive Correlations and statistically significant:

- First Serve Percentage (First_pct_rolling) and Second Serve Percentage (Second_pct_rolling): (0.88)
- This suggests that players who tend to have a high first serve percentage also tend to have a high second serve percentage. This indicates a general consistency in serving performance.
- Second Serve Percentage (Second_pct_rolling) and Forehand Winners (Winners_FH_rolling): (0.81)
This correlation might seem counterintuitive, but it could indicate that when a player is hitting their second serve well, they might be setting up more opportunities to dictate the point and hit forehand winners.
- Forehand Winners (Winners_FH_rolling) and Forehand Unforced Errors (UFE_FH_rolling): (0.89)
This is a common observation in tennis. Players who go for more winners on their forehand are also more likely to make unforced errors on that side. It's a trade-off between aggression and consistency.

Moderate Positive Correlations:

- First Serve Percentage (First_pct_rolling) and Forehand/Backhand Winners (Winners_FH/BH_rolling): (0.50, 0.55) This suggests a moderate relationship between first serve percentage and the ability to hit winners, indicating that a good first serve can create more opportunities for aggressive shots.
- Return Points Won Percentage (RPW_pct_rolling) and Backhand Winners (Winners_BH_rolling): (0.56) This indicates that players who are effective at returning serve also tend to hit more winners on their backhand side.

Strong Negative Correlations:

- Return Points Won Percentage (RPW_pct_rolling) and Backhand Unforced Errors (UFE_BH_rolling): (-0.72) This makes intuitive sense. Players who make fewer unforced errors on their backhand tend to win more return points. It highlights the importance of consistency in returning serve.
- First Serve Percentage (First_pct_rolling) and Backhand Unforced Errors (UFE_BH_rolling): (-0.40)
This suggests a moderate negative relationship between first serve percentage and making unforced errors on the backhand side, indicating that a good first serve can put pressure on the opponent and reduce their consistency.